

*Università degli Studi di Verona**Laurea Magistrale in Ingegneria e Scienze Informatiche*

Quantum Computing — AA 2025–2026

Exam Project P1

Quantum Desk Calculator

Reversible addition and subtraction of two positive integers

Student	Usama Ahmad
Student ID	VR513221
Supervisor	Prof. Alessandra Di Pierro
Implementation	Qiskit 2.x + Aer simulator
Language	Python 3

The circuit is implemented in Qiskit and demonstrated on two instances, as required by the exam project list.

1. Task

Project P1 asks to build quantum circuits that perform arithmetic operations between the binary representations of two positive integers. The circuit must be implemented in Qiskit and demonstrated on one or two instances. This report implements addition and subtraction on 3-bit unsigned integers (values 0 through 7) and demonstrates two instances of each operation.

2. Background

A quantum computation, in the circuit model of the course, starts in a computational-basis state, applies a sequence of one- and two-qubit unitary gates, and ends with a measurement in the computational basis. Because every gate is unitary, the evolution is reversible: information cannot be erased. Classical addition $a, b \mapsto a + b$ overwrites the inputs, so it is not a valid quantum operation. The reversible form used here is

$$|a\rangle |b\rangle |0\rangle \rightarrow |a\rangle |b\rangle |a + b\rangle$$

The inputs remain available after the computation. This is the standard Bennett-style embedding of irreversible arithmetic into a unitary circuit.

3. Reversible full adder

A one-bit full adder computes the sum bit and the outgoing carry from bits a, b and incoming carry c_{in} :

$$sum = a \oplus b \oplus c_{in}$$

$$c_{out} = majority(a, b, c_{in}) = (a \wedge b) \oplus (c_{in} \wedge (a \oplus b))$$

XOR is the CNOT gate (CX). AND is the Toffoli gate (CCX). The implementation copies $a \oplus b$ onto a scratch qubit so that a and b are never overwritten, writes the sum into a fresh bit, writes the carry into the next carry wire, and uncomputes the scratch qubit by applying the same CNOTs again. After uncomputation the scratch returns to $|0\rangle$ and can be reused at the next bit.

Chaining three full adders gives a ripple-carry adder. An extra sum qubit stores the last carry, so the output has 4 bits and can represent sums up to 14. That is enough for every pair of 3-bit positive integers, including $7 + 7 = 14$.

4. Subtraction

Subtraction is not a new circuit. For $a \geq b$ one uses two's complement:

$$a - b = a + (\sim b) + 1$$

Bitwise NOT is the Pauli-X gate on each bit of b . The extra +1 is an X gate on the initial carry. After the ripple adder, the lower 3 bits of the sum register are the difference. The X gates on b are applied again at the end so that register b is restored. The implementation assumes $a \geq b$, as required for a non-negative difference of two positive integers.

5. Qiskit implementation

The file `calculator.py` builds two circuits, `build_adder` and `build_subtractor`. Registers are:

Register	Qubits	Role
a	3	first integer, little-endian

b	3	second integer, little-endian
c	4	carry chain; c0 is the initial carry
sum / diff	4	result (4 bits for addition, 3 measured for subtraction)
scratch	1	temporary copy of $a \oplus b$

An integer v is loaded with X gates on those qubits whose bits are 1. The circuit is transpiled and executed on the Aer simulator with 1024 shots. Because the circuit is deterministic and the simulator is noiseless, every shot returns the same computational-basis string. Independently of the four demo instances, the same circuits were checked on every pair of 3-bit integers (all 64 additions, and all subtractions with $a \geq b$).

6. Demonstrated instances

The project brief requires one or two instances. Four instances are reported: two additions and two subtractions.

Operation	Decimal	Binary	Measured	Shots
addition	3 + 5	011 + 101	1000 = 8	1024 / 1024
addition	6 + 2	110 + 010	1000 = 8	1024 / 1024
subtraction	7 – 3	111 – 011	100 = 4	1024 / 1024
subtraction	5 – 1	101 – 001	100 = 4	1024 / 1024

The instance $3 + 5$ is chosen because it produces a carry out of the 3-bit range ($8 = 1000_2$), so the extra sum qubit is actually used. $6 + 2$ is a second independent pair with the same sum. $7 - 3$ and $5 - 1$ check two's-complement subtraction on a maximal 3-bit minuend and on a smaller pair.

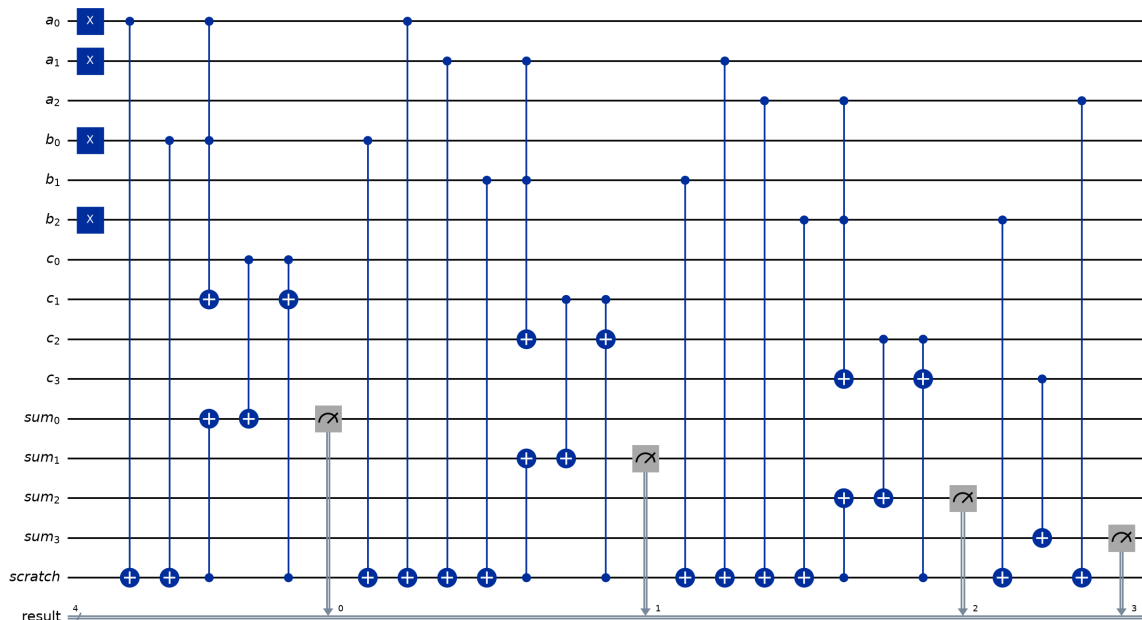


Figure 1. Qiskit circuit for $3 + 5$. X gates load 011 and 101; CNOT and Toffoli implement the ripple-carry adder; the four sum qubits are measured.

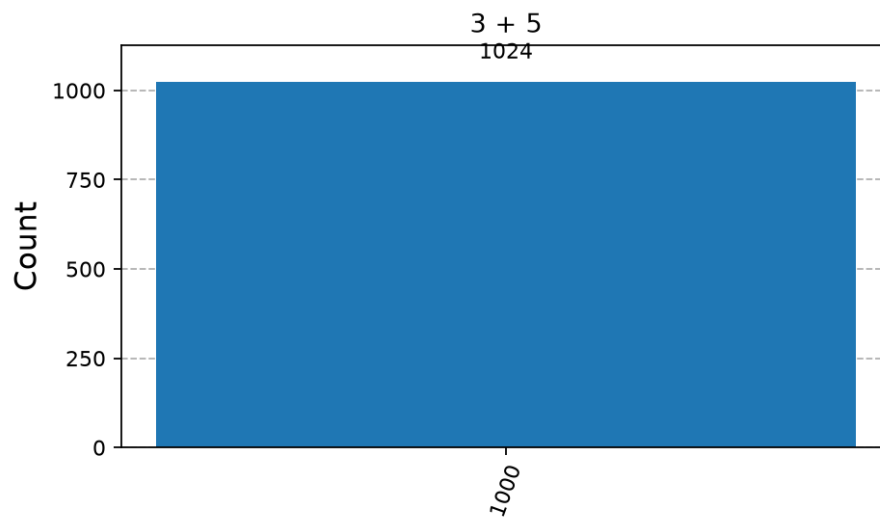


Figure 2. Measurement histogram for $3 + 5$ (Aer, 1024 shots). The only outcome is 1000, that is 8.

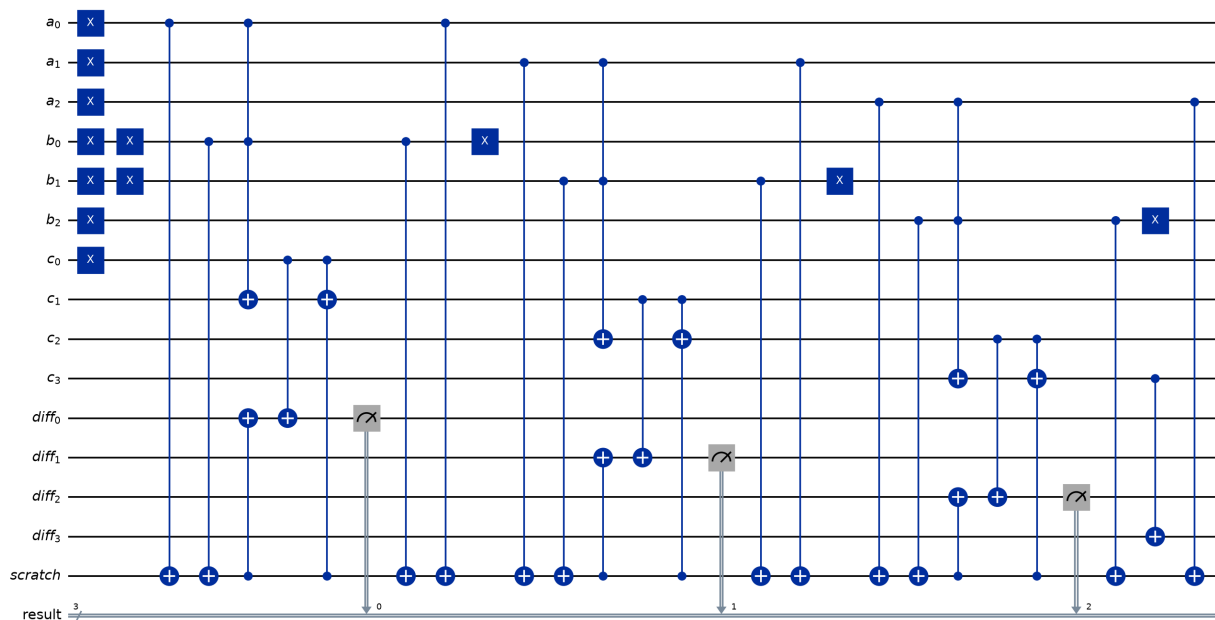


Figure 3. Qiskit circuit for $7 - 3$. X on c_0 injects the +1 of two's complement; X on each b qubit implements bitwise NOT.

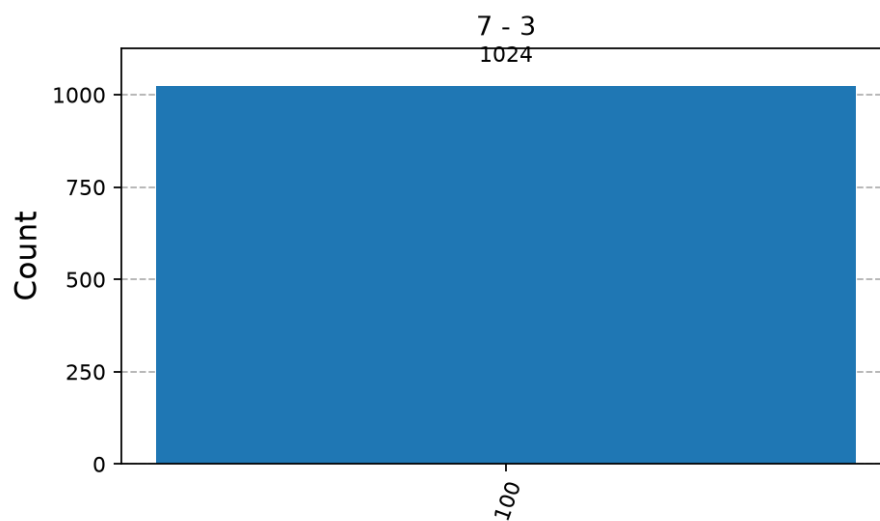


Figure 4. Measurement histogram for $7 - 3$. The only outcome is 100, that is 4.

7. How to run the demonstration

From the folder `p1-quantum-desk-calculator`:

```
pip install -r requirements.txt
python calculator.py
```

The Jupyter notebook `P1_Demo.ipynb` is the Qiskit demo: it draws the circuits, runs the four instances, and plots the histograms.

8. Scope and limitations

The circuits run on a noiseless simulator, which is sufficient for the required demonstration. They do not claim a computational speedup over a classical adder: the point of P1 is to realise arithmetic as a unitary, reversible circuit. The bit width is 3 so that the diagrams remain readable; the same ripple-carry construction extends to any width. Subtraction is defined for $a \geq b$ only. No hardware backend (IBM Quantum) was used.

9. Conclusions

A reversible quantum desk calculator for addition and subtraction of two positive integers was implemented in Qiskit. Each operation was demonstrated on two instances. In every case the Aer simulator returned the classical result on all shots, confirming that the circuits compute the intended arithmetic.

References

A. Di Pierro, *Quantum Computing*, lecture notes, University of Verona, AA 2025–2026. Project list: P1 Quantum Desk Calculator.

M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press. Reversible circuits and the Toffoli gate.

P. Kaye, R. Laflamme and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press. Circuit model.

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